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"A Unified Framework for Deep Multimodal Audio-Visual Steganography"

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CERTIFICATE

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In partial fulfilment for the completion of Fifth-semester Capstone Project Phase - 2 (UE23CS320B) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Jan. 2026 – May. 2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 6th-semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase - 1 entitled **“A Unified Framework for Deep Multimodal Audio Visual Steganography”** has been carried out by us under the guidance of **Dr. Surabhi Narayan** and submitted in partial fulfilment of the course requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Jan – May 2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

This Phase 2 report presents the implementation of a unified framework for deep multimodal audio-visual steganography, aimed at hiding a complete secret audio-video payload inside a cover audio-video stream while preserving imperceptibility, recoverability, and operational stability. The project extends the Phase 1 conceptual design into a concrete, modular system built around a shared temporal synchronization layer, where both audio and video are divided into fixed 1-second chunks using a common time-based chunk index. This synchronization strategy ensures that the audio and video streams remain aligned during both embedding and extraction, avoids gradual drift across the two modalities, and makes the encode-decode process deterministic even when the media undergoes common transformations such as trimming, remuxing, or re-encoding. The audio subsystem processes cover and secret audio after resampling them to 16 kHz and segmenting them into 16,000-sample chunks, then uses an invertible neural network (INN) to embed the secret signal while preserving the perceptual quality of the stego audio. The video subsystem follows a reversible hiding approach that uses temporal windowing, transform fusion, coupling-based invertible blocks, and redundancy estimation to hide secret video content inside the cover video with minimal visual distortion. By designing the audio and video branches as independent yet time-synchronized components, the framework supports chunk-wise hiding and recovery without requiring cross-chunk context, which simplifies implementation while maintaining strong control over alignment and reversibility. The system also incorporates loss functions and evaluation measures that balance cover quality, secret reconstruction quality, and robustness against compression-related distortions. Overall, this phase focuses on the finalized architecture, methodology, and module-level implementation of the proposed system, along with initial validation of stego quality, reconstruction fidelity, and runtime feasibility.

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CHAPTER 1

INTRODUCTION

1.1 Overview of Data Hiding and Steganography

Steganography is the practice of concealing the existence of information inside an ordinary-looking medium so that the hidden communication is not noticeable to an unintended observer. Unlike cryptography, which protects the content of a message by converting it into an unreadable form, steganography aims to hide the fact that communication is happening at all. In the digital age, where large volumes of audio, video, image, and text data are continuously shared across networks, steganography has become an important technique for secure and covert communication. Traditional data hiding methods were mainly based on fixed rules such as least significant bit manipulation, but these approaches often suffer from low capacity, poor robustness, and vulnerability to compression or detection. To overcome these limitations, modern steganography increasingly uses deep learning methods that can learn complex hiding and recovery strategies from data.

1.2 Motivation for Multimodal Audio-Visual Steganography

Most conventional steganography systems operate on a single medium such as only audio or only video. However, a single modality often has limited payload capacity and is highly sensitive to distortion. Audio steganography is constrained by the Human Auditory System, which is very sensitive to small waveform changes, while video steganography must preserve visual quality to avoid visible artifacts. Because of these limitations, hiding a large payload such as a complete secret video becomes difficult using only one channel. Multimodal steganography addresses this problem by using both the visual and audio components of a video as carriers. This provides higher capacity,

improved security, and better flexibility in distributing the secret payload across different streams. The motivation behind this project is to build a system that can jointly support audio and video hiding while keeping the final stego media perceptually close to the original cover media.

1.3 Need for a Unified Framework

Although multimodal hiding is powerful, simply using separate audio and video steganography pipelines is not sufficient for reliable recovery. If the two streams are processed independently without a common alignment strategy, the hidden information can drift over time and decoding may become inconsistent. To solve this, the proposed system uses a shared temporal synchronization layer that divides both audio and video into fixed 1-second chunks using a common time-based index. This makes the encode and decode process deterministic and ensures that corresponding audio and video segments always refer to the same real-time interval. The phase 2 design uses this synchronization layer as the central binding mechanism between the audio and video branches, so that both modalities remain aligned throughout hiding and extraction.

1.4 Project Focus

The project titled **“A Unified Framework for Deep Multimodal Audio-Visual Steganography”** focuses on implementing a reversible, chunk-based system for hiding a secret audio-video payload inside a cover audio-video stream. The audio branch uses an invertible neural network based embedding pipeline with 16 kHz resampled audio and fixed 1-second chunks, while the video branch uses a reversible hiding core with temporal windowing, transform fusion, and redundancy estimation. The overall framework is modular, allowing audio and video processing to happen independently while still remaining synchronized through the same chunk index. This phase of the project emphasizes the complete system architecture, methodology, implementation of major modules, and evaluation of stego quality, secret recovery quality, and runtime feasibility.

1.5 Scope of the Work

The scope of this phase includes the design and implementation of the temporal synchronization module, the audio-in-audio steganography INN, and the video-in-video steganography INN. The system processes the media in fixed-size time chunks, embeds the secret content in a reversible manner, and reconstructs the hidden payload during decoding using the same chunk boundaries. The work also includes preprocessing, post-processing, and evaluation of cover fidelity and secret reconstruction quality. By using a synchronized and modular design, the project aims to provide a stable foundation for secure multimodal data hiding and future extensions toward stronger robustness and improved joint embedding strategies.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of Deep Learning Steganography

Traditional steganography techniques relied on fixed rule-based methods such as least significant bit modification, which are simple to implement but limited in payload capacity, robustness, and resistance to detection. With the growth of multimedia communication, deep learning-based steganography has emerged as a stronger alternative because neural networks can learn complex hiding and recovery mappings directly from data. In particular, reversible and invertible neural network architectures have gained importance because they allow the hidden content to be recovered with minimal loss while also preserving the quality of the cover media. For audio-visual steganography, this is especially valuable because the system must maintain both visual imperceptibility and acoustic fidelity while also supporting a sufficiently large hidden payload.

2.2 Hide and Speak: Deep Neural Networks for Speech Steganography

Hide and Speak is a deep neural network approach for audio steganography proposed by Kreuk et al. The method was designed to address the limitations of conventional audio hiding techniques, which often introduce audible artifacts and fail to preserve waveform quality. A key contribution of this work is the use of differentiable STFT and iSTFT layers inside the network, which forces the model to preserve audio fidelity during both embedding and recovery. The architecture consists of an encoder network, a carrier decoder network, and a message decoder network, with gated convolutional layers used for feature extraction. The method also supports hiding multiple messages in a single carrier signal. The reported results show strong imperceptibility and robustness, with

audio quality close to the original signal and resilience against distortions such as MP3 compression, noise, downsampling, and filtering. This work is highly relevant to the audio branch of our project because it shows that reversible audio hiding can be achieved only when the model is trained with strong fidelity constraints in both the time and frequency domains.

2.3 Large-Capacity and Flexible Video Steganography via Invertible Neural Network (LF-VSN)

LF-VSN is an invertible neural network-based method for video steganography proposed by Mou et al. It addresses the problem of limited capacity in existing video hiding systems by using a reversible architecture that can hide and recover multiple secret videos inside a single cover video. The method uses a wavelet-based transformation to separate frequency components and then applies invertible blocks that use additive and affine coupling layers for reversible information flow. A redundancy prediction module is used to estimate missing latent information instead of relying on random sampling, improving recovery stability. The reported results demonstrate high PSNR and SSIM values even when multiple videos are embedded, along with strong security against steganalysis. This work is important for our project because it provides a practical example of reversible video hiding using invertible neural networks, and it motivates the use of temporal windowing and redundancy estimation in the video branch.

2.4 Reversible Audio Steganography with WavInWav

The WavInWav approach presents an invertible neural network framework for audio-in-audio steganography. In this method, both cover and secret audio are resampled to a fixed rate of 16 kHz and divided into 1-second chunks of 16,000 samples. These chunks are then passed through a sequence of invertible blocks containing coupling layers, affine transformations, and channel shuffling. The network generates a stego audio output along with an auxiliary latent variable that captures the leftover information required for exact recovery. During training, a time-frequency loss

is applied to reduce audible artifacts and preserve both waveform and spectral quality. The main idea behind this approach is that scaling and shifting operations can be reversed exactly if the same parameters are available during decoding. This makes WavInWav a strong reference for our audio module, especially because our system also relies on fixed-length chunking, invertible transformations, and frequency-aware loss design.

2.5 Research Gap and Relevance to the Proposed Work

From the reviewed literature, it is clear that audio steganography and video steganography have each made strong progress individually, especially through invertible neural networks and frequency-aware loss functions. However, most existing systems still treat audio and video as separate problems rather than as a unified multimodal hiding framework. Existing methods also tend to focus on either high capacity, imperceptibility, or robustness, but not all three together in a synchronized audio-visual setting. Based on this gap, the proposed project combines a reversible audio branch, a reversible video branch, and a shared temporal synchronization layer so that secret audio and secret video can be embedded and recovered in a coordinated way. This makes the system more suitable for high-capacity multimedia steganography and provides a strong base for further improvements in robustness and joint encoding.

CHAPTER 3

ARCHITECTURE WITH EXPLANATION

3.1 Overall System Architecture

The proposed system, “A Unified Framework for Deep Multimodal Audio-Visual Steganography,” is designed as a modular framework for hiding a secret audio-video payload inside a cover audio-video stream. The overall architecture is divided into two major flows: encoding and decoding. During encoding, the secret and cover media are separated, processed, and combined to produce a stego output. During decoding, the stego container is separated again and the hidden secret media is recovered. The main architectural goal is to keep the audio and video streams synchronized while allowing reversible embedding and extraction.

The architecture shown, places the Temporal Synchronization Module between input preprocessing and the two embedding branches, and it uses separate Video Steganography INN and Audio Steganography INN blocks before final post-processing. The decoder follows the same modular structure in reverse, using separation, temporal realignment, parallel decoding, and recombination. This makes the system easy to understand, modular to implement, and consistent across encoding and decoding.

3.2 Encoder Architecture

The encoder takes secret AV input and cover AV input as separate sources. These inputs are first demultiplexed into their audio and video components. The separated streams are then passed through the temporal synchronization stage, which ensures that the audio and video segments correspond to the same time interval. After synchronization, the aligned streams move into two parallel embedding

paths: one for video and one for audio. The outputs of these two paths are finally combined through post-processing to generate the stego video output.

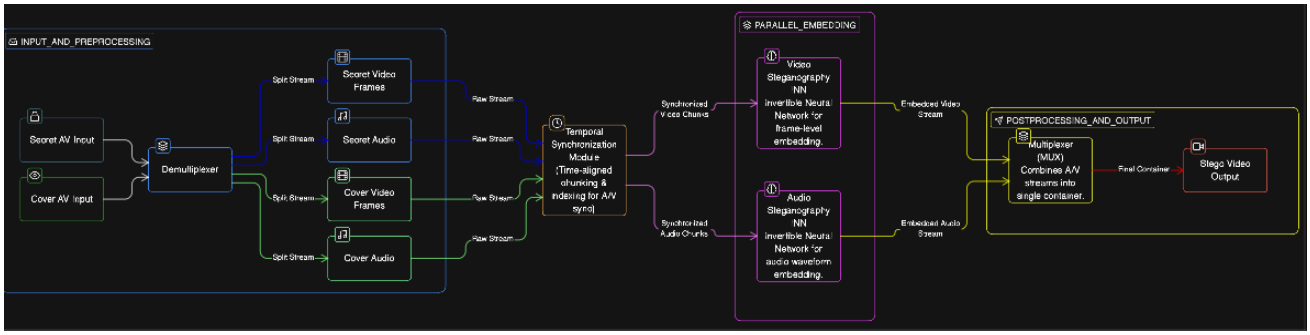


Figure 3.1: Encoder Architecture Diagram

In the encoder diagram, the main architectural blocks are clearly visible: Input and Preprocessing, Temporal Synchronization, Parallel Embedding, and Postprocessing and Output. This structure shows that the system is not a single monolithic model, but a coordinated pipeline where each component has a specific role. The architecture supports clear separation of responsibilities and makes the flow from input media to stego container easy to trace.

3.3 Decoder Architecture

The decoder starts with the stego video input, which is first separated into audio and video streams using a demultiplexer. The separated streams are then realigned using the same temporal indexing logic used during encoding. After realignment, the streams are passed through the corresponding decoding branches to recover the hidden video and audio content. Finally, the recovered streams are recombined to produce the recovered secret AV.

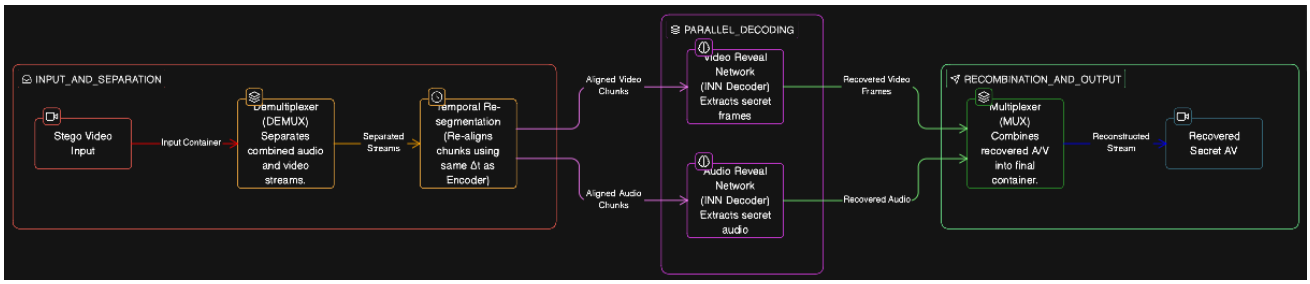


Figure 3.2: Decoder Architecture Diagram

The decoder diagram follows the same modular structure as the encoder, which is important because the entire system depends on symmetry between the forward and reverse paths. The architecture therefore makes the extraction process predictable and consistent. Since the same alignment principle is used in both directions, the decoder can reconstruct the hidden payload from the stego container in a structured way.

3.4 Architectural Summary

In summary, the proposed framework uses a two-stage reversible architecture with separate encoding and decoding pipelines, linked by a shared synchronization mechanism. The main components are the demultiplexer, temporal synchronization block, parallel audio/video embedding blocks, and final recombination stage. This architecture provides a clear and modular base for the implementation of multimodal steganography in Phase 2.

CHAPTER 4

METHODOLOGY

4.1 Overview of the Methodology

The proposed system follows a modular, chunk-based methodology for reversible multimodal steganography. The main idea is to process the cover and secret audio-video streams in synchronized time intervals so that the corresponding audio and video segments always represent the same real-time portion of the media. This is achieved through a shared Sync Map, which divides both modalities into fixed 1-second chunks and keeps the encoder and decoder aligned using the same chunk indexing logic. The methodology is designed so that audio and video can be embedded independently while still remaining temporally coherent as a single framework.

4.2 Temporal Synchronization Method

The first step in the methodology is temporal synchronization. The cover and secret media are separated into audio and video streams, and then a common time grid is created. Each chunk corresponds to the time interval from k to $k + 1$seconds. For audio, the signals are resampled to 16 kHz, and each chunk contains exactly 16,000 samples. For video, frames are assigned to chunks using timestamp-based grouping, where the chunk ID is determined by the floor of the frame timestamp. This ensures that audio chunk k and video chunk k correspond to the same time interval.

This method is used because frame-count-based or sample-count-based slicing can create drift when frame rates vary or when the media is trimmed or re-encoded. By using a deterministic timestamp-based Sync Map, the system avoids misalignment during both encoding and decoding. The same chunk boundaries are used in the reverse path, which makes the overall process stable and reversible.

4.3 Audio Embedding Method

The audio subsystem uses an invertible neural network based pipeline for hiding the secret audio inside the cover audio. After resampling and chunking, both cover and secret audio segments are normalized and passed as paired inputs to the Audio-INN module. The network processes each chunk through a sequence of invertible blocks that use coupling layers, affine transformations, and channel shuffling. These operations mix the secret content into the cover signal while still preserving the ability to reverse the transformation later.

The audio module produces two outputs: the stego audio chunk and an auxiliary latent variable that stores the information needed for exact reconstruction. During training, the model uses a time-frequency loss, which compares the stego audio with the cover audio in both the raw sample domain and the frequency domain. This helps reduce audible artifacts and keeps the hidden audio perceptually close to the original cover. The reverse pass uses the same invertible structure to recover the secret audio from the stego audio.

4.4 Video Embedding Method

The video subsystem follows a reversible hiding strategy similar in spirit to the audio branch, but adapted for frame-level processing. The video stream is divided into synchronized chunks, and a sliding temporal window is used to provide local context for embedding. The **Video-INN** module processes the cover and secret video chunks using a transform fusion stage and invertible coupling blocks. This helps the model embed the secret frames while keeping the visible differences between the cover and stego frames minimal.

The method includes a **redundancy estimator**, which predicts the latent information required for the reverse pass instead of depending on random sampling. This makes secret recovery more stable. The

architecture notes also describe a decoder broadcast strategy for handling the stego frame during extraction. Overall, the video methodology is designed to support reversible frame-level embedding while maintaining high visual quality.

4.5 Encoding and Decoding Procedure

During encoding, the secret AV input and cover AV input are first separated into audio and video streams. The streams are synchronized using the common time-based chunk index. After synchronization, the aligned audio chunks are passed through the Audio-INN branch, and the aligned video chunks are passed through the Video-INN branch. The embedded audio and video outputs are then recombined using post-processing to generate the final stego container.

During decoding, the stego AV input is separated again into audio and video streams. The same chunk indexing logic is applied to restore the original alignment. The video decoder extracts the hidden video frames, and the audio decoder reconstructs the secret audio chunks. The recovered outputs are finally combined to reconstruct the hidden secret AV stream. Because the same deterministic chunk boundaries are used in both directions, the system supports reversible recovery with minimal drift.

4.6 Design Constraints and Assumptions

The methodology is built around a few fixed design constraints. The chunk duration is set to **1 second**, the audio sample rate is fixed at **16 kHz**, and the audio chunk size is fixed at **16,000 samples**. The system assumes that the cover and secret media are time-aligned and that video timestamps are available for deterministic grouping. It also assumes that encoder and decoder use the same chunk indexing logic, since any mismatch would break recovery. These constraints simplify the implementation and make the framework more stable for reversible steganography.

CHAPTER 5

IMPLEMENTATION

5.1 Overview of Implementation

The proposed system is implemented as a modular pipeline where each component is responsible for a specific stage of the steganography process. The implementation follows the architecture described in Chapter 3 and the methodology outlined in Chapter 4, focusing on chunk-based processing, synchronization, and reversible embedding. The system is divided into preprocessing, synchronization, audio embedding, video embedding, and post-processing modules. Each module operates independently but is coordinated through a shared time-based indexing mechanism to ensure consistency during encoding and decoding.

5.2 AV Preprocessing Module

The preprocessing module is responsible for preparing the input data before embedding. The input cover and secret video files are first demultiplexed into separate audio and video streams. The audio signals are resampled to a fixed rate of 16 kHz to maintain consistency across all inputs. Video frames are extracted along with their timestamps to enable time-based grouping.

After separation, both audio and video streams are normalized and formatted into structures suitable for chunk-based processing. This module ensures that the inputs are clean, consistent, and compatible with the subsequent synchronization and embedding stages.

5.3 Temporal Synchronization Module (Sync Map)

The Temporal Synchronization Module is a core component of the implementation. It constructs a shared Sync Map, which divides both audio and video streams into fixed 1-second intervals. For audio, this corresponds to segments of 16,000 samples, while for video, frames are grouped based on their timestamps using integer-based time intervals.

Each chunk is assigned a global index *k*, ensuring that audio chunk *k* and video chunk *k* correspond to the same real-time segment. This indexing is used consistently during both encoding and decoding. The implementation ensures deterministic slicing, which prevents drift and guarantees that the same chunk boundaries are used during recovery.

5.4 Audio Steganography Module (Audio-INN)

The audio steganography module is implemented using an Invertible Neural Network (INN) that embeds secret audio into cover audio on a per-chunk basis. Each pair of cover and secret audio chunks is passed through a sequence of invertible blocks consisting of coupling layers, affine transformations, and channel shuffling.

The forward pass generates the stego audio chunk, which closely resembles the cover audio, along with an auxiliary latent variable used for reversibility. During training, a time-frequency loss is applied to ensure that the embedded audio maintains both waveform similarity and spectral fidelity.

During decoding, the same INN structure is used in reverse to reconstruct the secret audio from the stego audio. The implementation ensures that the forward and reverse transformations use identical parameters, enabling accurate recovery.

5.5 Video Steganography Module (Video-INN)

The video steganography module is implemented as a reversible hiding system based on invertible transformations. The video stream is processed in synchronized chunks, and a sliding window is used to provide local temporal context. For each time step, a group of frames is taken as input, and the model produces a single stego frame corresponding to the central time index.

The module includes a transform fusion stage to combine cover and secret features and a series of invertible coupling blocks to embed the secret information. A redundancy estimator is used to predict latent information required for the reverse pass, improving stability during decoding.

During extraction, the stego frames are processed through the reverse pipeline to reconstruct the secret video frames. The implementation ensures that temporal consistency is maintained across frames within each chunk.

5.6 AV Post-processing Module

The post-processing module is responsible for combining the processed audio and video streams into a final output format. After embedding, the stego audio chunks are concatenated to form a continuous audio stream, and the stego video frames are combined to reconstruct the video stream.

These streams are then multiplexed into a standard container format to produce the final stego AV file. During decoding, the reverse process is applied to reconstruct the recovered secret audio and video streams from their respective chunks.

CHAPTER 6

RESULTS

6.1 Overview

This chapter presents the evaluation of the proposed multimodal steganography system based on the implemented audio and video pipelines. The results focus on three key aspects:

1. Imperceptibility of the stego output
2. Quality of secret reconstruction
3. Feasibility of the implementation

The evaluation is performed separately for the video and audio branches using standard quality metrics.

6.2 Video Steganography Results

The performance of the video steganography module is evaluated using PSNR (Peak Signal-to-Noise Ratio) and SSIM (Structural Similarity Index). These metrics measure the similarity between the cover, stego, and recovered secret video frames.

Test Configuration:

- Test samples: 200
- Model checkpoint: trained Video-INN model

Results:

Metric	Value
PSNR (Cover)	22.6197 dB
PSNR (Secret)	22.5737 dB
PSNR (Stego)	22.5418 dB
SSIM (Cover)	0.7103
SSIM (Secret)	0.7069
SSIM (Stego)	0.6857

Table 6.1: Video Steganography Result Metrics

Analysis

The PSNR values for cover, secret, and stego are all around 22.5 dB, indicating that the stego frames remain close to the original cover frames while still embedding the secret information. The SSIM values are also consistent across all three categories, remaining close to 0.7, which suggests that structural similarity is largely preserved.

The small difference between cover and stego metrics shows that the embedding process introduces only limited distortion. Similarly, the secret reconstruction maintains comparable quality, indicating that the reversible architecture is functioning correctly. These results demonstrate a balanced trade-off between embedding capacity and visual quality.

6.3 Audio Steganography Results

The performance of the audio steganography module is evaluated using **Signal-to-Noise Ratio (SNR)**, which measures how well the secret audio is preserved after embedding and recovery.

Results:

Data Type	Secret SNR
float32	137 dB
float16	74 dB
int8	43 dB

Table 6.2: Audio Steganography Result Metrics

Analysis

The results show that the system achieves very high reconstruction quality for audio. In the float32 configuration, the SNR reaches approximately 137 dB, which indicates near-perfect recovery of the secret audio. Even with reduced precision formats such as float16 and int8, the system maintains strong performance with SNR values of 74 dB and 43 dB, respectively.

These results demonstrate that the audio embedding process is highly effective and robust to different data representations. The use of invertible neural networks and time-frequency loss contributes to preserving both waveform integrity and perceptual quality.

6.4 Discussion

The experimental results highlight the effectiveness of the proposed multimodal framework. The video module achieves stable reconstruction with consistent PSNR and SSIM values, while the audio module demonstrates extremely high reconstruction accuracy, especially at higher precision levels.

One important observation is that the system maintains a balance between imperceptibility and recoverability. The stego outputs remain visually and acoustically close to the original cover media, while the hidden secret can still be reconstructed with good fidelity.

The results also validate the design choice of using chunk-based synchronization, as the system is able to process and reconstruct data consistently across both modalities without noticeable misalignment.

6.5 Limitations

Despite the promising results, there are some limitations:

- The video PSNR and SSIM values indicate moderate visual quality and can be further improved.
- The system currently processes data in fixed 1-second chunks, which limits long-range temporal modeling.
- Evaluation is performed on a limited dataset and may require broader testing for generalization.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

This work presented the implementation of a unified multimodal audio-visual steganography framework that embeds a secret audio-video payload within a cover audio-video stream using reversible neural networks. The system was designed as a modular pipeline with separate audio and video branches, coordinated through a shared time-based synchronization mechanism.

The use of 1-second synchronized chunking ensured consistent alignment between audio and video during both encoding and decoding, preventing temporal drift. The Audio-INN module demonstrated very high reconstruction quality, while the Video-INN module achieved stable embedding with acceptable visual fidelity. Overall, the system successfully balances imperceptibility and recoverability, validating the feasibility of chunk-based reversible multimodal steganography.

7.2 Future Work

Future improvements can focus on enhancing both performance and robustness. The visual quality of the video branch can be improved using deeper architectures or better perceptual loss functions. Extending the system to support joint audio-video encoding instead of parallel independent branches could further improve efficiency and security.

Additionally, robustness against real-world distortions such as compression, noise, and frame rate changes can be strengthened. Incorporating longer temporal context beyond fixed 1-second chunks and optimizing the system for faster inference are also important directions for future work.

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